MUHAMMAD ABBAS HUSSAIN

Senior Game Developer | Gameplay & Multiplayer Programming | Unreal Engine 5 | Unity

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PROFESSIONAL SUMMARY

Senior Game Developer with 6+ years of experience building gameplay, multiplayer and VR products in Unreal Engine 5 and Unity. Specializes in C++ gameplay programming, network replication, AI behavior trees and engine plugin development. Founder of Different Slice, an indie game studio that also builds real-time AI MetaHuman assistants for clients. Has delivered for remote teams in the UK, Ireland and the US.

SKILLS

Engines: Unreal Engine 5, Unreal Engine 4, Unity (2019 to Unity 6), Unity WebGL

Languages: C++, C#, Blueprints, Python, JavaScript

Gameplay Systems: AI Behavior Trees, Animation Systems, Inverse Kinematics (Full Body IK, FABRIK), Physics, UI (UMG, Slate)

Multiplayer: Network Replication, Steamworks, Epic Online Services (EOS), Photon PUN 2, Matchmaking, UPnP / Port Forwarding, WebSockets

Optimization: Rendering and LOD Optimization, Replication Optimization, VR Frame-Rate Targets (90 fps)

VR / XR: Meta Quest (Oculus SDK), VR Interaction, Motion Controllers

Platforms: Windows PC, Android, iOS, Meta Quest, WebGL

Tools: Perforce, Git / GitHub, Jira, Trello, Visual Studio, JetBrains Rider, Agile / Scrum

PROFESSIONAL EXPERIENCE

Founder & Lead Developer **2026 – Present**

Different Slice | *Remote* | differentslice.com

- Founded an independent game studio that develops games and offers real-time development services to clients, working across engines.
- Lead engineering on client products, including a real-time AI MetaHuman assistant and an interactive UE5 interior configurator with live furniture, material and lighting changes.

Senior Game Developer **Feb 2025 – Present**

4BitLab | *Manchester, UK (Remote)*

- Architected a UE5 C++ networking plugin using UPnP and port forwarding for direct server-client connections without a relay service, cutting connection setup time by 40%.
- Led gameplay development of a multiplayer FPS, building weapon systems, movement controllers and Steam matchmaking.
- Designed AI behavior trees with adaptive difficulty that scales to player performance.
- Raised performance on mid-tier hardware by 30% by optimizing the rendering pipeline, LODs and network replication.
- Integrated Steamworks achievements, leaderboards and cloud save.

Senior Game Developer (Contract) **Jul 2024 – Sep 2024**

Magic Media | *Dublin, Ireland (Remote)*

- Built core gameplay for Astral Sever (formerly Revenge), a UE5 first-person multiplayer shooter coming to Steam: player movement, weapon handling and combat.
- Implemented modular C++ and Blueprint systems for player abilities and dynamic level interactions.
- Worked in a 6-developer team on Perforce, keeping clean changelists and reviewing teammates' code.

Game Developer **Jul 2022 – Jan 2025**

QBit Technologies | *Palo Alto, CA, USA (Remote)*

- Built VR interaction systems for Unity applications in historical simulation, industrial training and a blockchain metaverse, holding 90 fps on Meta Quest.
- Created branching dialogue with real-time voice acting for The August Choice, a historical VR RPG set in Ancient Rome.
- Delivered a 1:1-scale enterprise VR training simulator for server component replacement, with physical interactions (bolts, switches, forklift) and assessment tracking.
- Integrated a Polygon token economy and smart contracts into VR City, a metaverse with avatar customization, land ownership and in-world retail.

Junior Game Developer **Mar 2020 – Apr 2022**

Frag Games | *Lahore, Pakistan*

- Built gameplay for Boddle Learning, a 3D Unity EdTech game used in U.S. classrooms.
- Ported Asterix and Friends, a Flash multiplayer browser game, to Unity WebGL, fixing rendering and input compatibility across browsers.
- Built and iterated on hyper-casual Unity mobile prototypes in short cycles, focused on retention and monetization KPIs.
- Contributed gameplay integration and debugging to MyWhoosh, an Unreal Engine 4 cycling simulation app.

PROJECTS

AI MetaHuman Assistant | *Unreal Engine 5, C++, MetaHuman, UMG, WebSockets, LLM* | differentslice.com

- Built a photoreal MetaHuman kiosk assistant in C++ that listens, speaks and answers in real time, with voice capture, real-time lip sync and a native UMG interface.
- Streamed audio over authenticated WebSockets with automatic reconnects, keeping the LLM backend swappable (ChatGPT, Gemini or custom).
- Made lighting, camera, UI layout and languages (including right-to-left) config-driven, with a web control page for live edits while the app runs.

Boddle Learning | *Unity, C#, EdTech* | boddlelearning.com

- Designed arcade-style minigames that reward students with gameplay power-ups for answering questions correctly.
- Built the gameplay loop, student progression tracking and integration with the question (QA) modules.
- Deployed in U.S. K-12 classrooms, reaching 1,000+ students.

Interior Configurator | *Unreal Engine 5, UMG* | differentslice.com

- Built an interactive house walkthrough: clicking furniture opens a product card with variants, and viewers swap materials, colors, wall finishes and time-of-day lighting presets live.

FitPicSimulator | *Unreal Engine 5, Full Body IK, FABRIK, Blueprints*

- Built an interactive posing system where users drag on-screen handles to move limbs, blending idle and user-set poses without breaking skeletal constraints.

Selected Titles: Astral Sever (Steam), MyWhoosh, Asterix and Friends, Bullet Rush (Google Play), Haemaccel Run (Google Play)

EDUCATION

Bachelor of Science in Computer Science **2016 – 2020**

National University of Computer and Emerging Sciences (FAST-NUCES)

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Graphics, Artificial Intelligence, Operating Systems